

M.Tech/PHD

Roll No.....

MID TERM EXAMINATION

Mar-2025

Pattern Recognition (AI504/ RICV502/RIRA502)

Time: 1:30 Hours

Max. Marks: 20

Note: All questions are compulsory.

Q.1 a) Explain the process of k-fold cross-validation and its role in preventing overfitting. [CO1] [2M]

b) Given a dataset with high dimensionality and non-linear relationships, which classifier SVM or KNN would you choose and why? Justify your choice. [CO1][2M]

c) Compare DBSCAN with Hierarchical Clustering. Under what conditions would you prefer one over the other? [CO2][2M]

Q 2) Given the following dataset, perform k-Means clustering using $k=2$. Assume the initial centroids are $C1=(2,2)$, and $C2=(6,6)$

Data points: (1,1), (2,3), (5,7), (6,5), (8,8). Assign each point to the nearest centroid and update the centroids. Show all steps. [CO2][3M]

Q 4) Consider a dataset with the following five points in 2D space:

A (2,3), B(5,6), C(8,8), D(3,5), E(7,7)

Compute the Euclidean distance matrix between all pairs of points. Perform Hierarchical Agglomerative Clustering (HAC). Draw the dendrogram showing how the clusters merge at each step. If we want two clusters, at what threshold distance should we cut the dendrogram? [CO2] [5M]

Q.5 You are given a 3D dataset with three variables:

$$X = \begin{bmatrix} 2 & 4 & 6 \\ 3 & 5 & 7 \\ 4 & 6 & 8 \\ 5 & 7 & 9 \end{bmatrix}$$

Perform the following tasks

- Compute the mean vector for each feature.
- Compute the covariance matrix of the dataset.
- Find the eigenvalues and eigenvectors of the covariance matrix.
- Identify the principal components and determine the most significant direction of variance.

[CO2][6M]